



Patterns

Nested Loops ✓

Loop inside a loop

1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4



Nested Loops

Loop inside a loop

	C1	C2	C3	C4
R1	1	1	1	1
R2	2	2	2	2
R3	3	3	3	3
R4	4	4	4	4

```
for (int i=1; i<=4; i++) {  
    for (int j=1; j<=4; j++) {  
        cout << i << " " << j << " ";  
    }  
}
```

① Outer loop : no. of Rows

② Inner loop : no. of columns/
each row

③ Work in inner loop (each row)
i print



code.cpp

code.cpp > main()

```

3
4  int main() {
5      //outer loop
6      for(int i=1; i<=4; i++){
7          //inner loop
8          for(int j=1; j<=4; j++){
9              //work
10             cout << i << " ";
11         }
12         cout << endl;
13     }
14     return 0;
15 }
```

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PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```

● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
1 1 1 1 2 2 2 2 3 3 3 3 4 4 4 4 %
● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
○ apnacollege@Amans-MacBook-Pro Cpp Codes %
```



Nested Loops

Dry Run

outer

i=1
val=1

inner

j=~~1~~ 2 3 4 5

i=2
val=2

j=~~1~~ 2 3 4 5

i=3
val=3

j=~~1~~ 2 3 4 5

i=4
val=4

j=~~1~~ 2 3 4 5

i=5

```
for(int i=1; i<=4; i++) {  
    → int val = i;  
    → for(int j=1; j<=4; j++) {  
        cout << val << " ";  
    }  
    cout << endl;  
}
```

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output

1 1 1 1

2 2 2 2

3 3 3 3

4 4 4 4



code.cpp ×

code.cpp > main()

```
5   int n = 4;
6   int i;
7   //outer loop
8   for(i=1; i<=n; i++) {
9       int val = i;
10      //inner loop
11      for(int j=1; j<=n; j++) {
12          //work
13          cout << val << " ";
14      }
15      cout << endl;
16  }
17
18  cout << i << endl;
19  return 0;
```

PORTS

PROBLEMS

OUTPUT

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```
● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
```

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
```

5

```
○ apnacollege@Amans-MacBook-Pro Cpp Codes %
```



code.cpp ×

code.cpp > main()

```
5     int n = 4;
6     int i;
7     //outer loop
8     for(i=1; i<=n; i++) {
9         int val = i;
10        //inner loop
11        for(int j=1; j<=n+1; j++) {
12            //work
13            cout << val << " ";
14        }
15        cout << endl;
```

PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

zsh

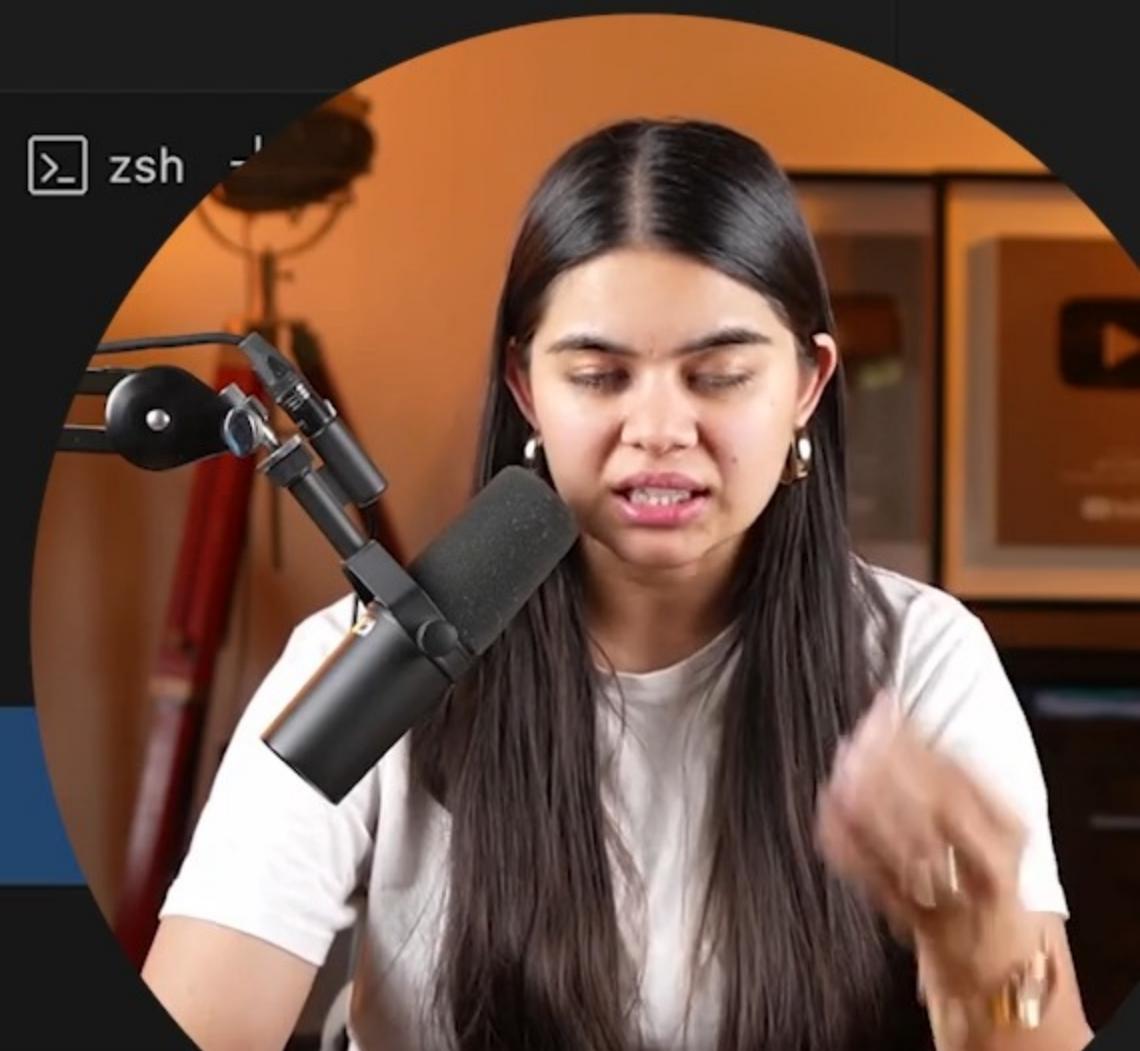
● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
5
```

● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
5
```

○ apnacollege@Amans-MacBook-Pro Cpp Codes %



Print Star pattern

n = 4

① outer loop $\rightarrow$ Rows

```
*  
* *  
* * *  
* * * *
```

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n = 4

Print Star pattern

n = 4

R1 * 1st
R2 ** 2st
R3 *** 3st
R4 **** 4st

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```
for (int i = 1; i <= n; i++) {  
    for (int j = 1; j <= i; j++) {  
        cout << "*" << " ";  
    }
```

```
    cout << endl;  
}
```

n = 4

}

① outer loop → Rows (i) → **APNA COLLEGE**
n times (1 to n)

② inner loop (each row's)
columns
i times (1 to i)

③ work ?

cout << "*" << " "



code.cpp

code.cpp > main()

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n = 4;
6
7      for(int i=1; i<=n; i++) {
8          for(int j=1; j<=i; j++) {
9              cout << "* ";
10         }
11
12         cout << endl;
13     }
14     return 0;
```

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PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
**
***
****
```

apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out

```
*
* *
* * *
* * * *
```

apnacollege@Amans-MacBook-Pro Cpp Codes %



Print Inverted Star pattern

```
* * * *  
* * *  
* *  
*
```

$n = 4$

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Print Inverted Star pattern

$n = 4$
 $R_1 = \text{*****}$ $4 = 4 - i + 1$
 $R_2 = \text{***}$ 3
 $R_3 = \text{**}$ 2
 $R_4 = \text{*}$ 1

① outer loop (rows)
(1 to n)

② inner loop (each row)

$2 \times i - 1$
 i
 $n - i$
 $n - i + 1$
 $n - i - 1$
 $n = 4$



Print Inverted Star pattern

$n = 4$

R1	*	*	*	*
R2	*	*	*	
R3	*	*		
R4	*			

$4 = n - i + 1$
 $3 = n - i + 1$
 $2 = n - i + 1$
 $1 = n - i + 1$

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① outer loop (rows)
(1 to n)

② inner loop (each row)
(1 to $n - i + 1$)

③ work?
cout << "*" << " "



n = 4

Print Half Pyramid pattern

1
12
123
1234

$n=4$
 $n=5$

|

$n = 4$



Print Half Pyramid pattern

$n=4$

R1 **1** 1
R2 **12** 1 to 2
R3 **123** 1 to 3
R4 **1234** 1 to 4

1 to i

```
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        cout << j << " ";
    }
    cout << endl;
}
```

}
end line

}

① outer loop (rows)
(1 to n)

② inner loop (each row)
(1 to i) $\Rightarrow j$

③ work?

cout << j << " ";



$n = 4$

code.cpp

code.cpp > main()

```
1  using namespace std;
2
3
4  int main() {
5      int n = 4;
6
7      for(int i=1; i<=n; i++) {
8          for(int j=1; j<=i; j++) {
9              cout << j;
10             }
11             cout << endl;
12         }
13         return 0;
14     }
```

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PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out

```
1
12
123
1234
```

apnacollege@Amans-MacBook-Pro Cpp Codes %



Print Character Pyramid pattern

A
BC
DEF
GHIJ

n = 4



Print Character Pyramid pattern

A
BC
DEF
GHIJ

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$n = 4$

char ch = 'A'

① outer loop (row)
(1 to n)

② inner loop (each row)
(1 to i)

③ work?

cout << ch;
ch++;



$n = 4$

Print Hollow Rectangle pattern

R1	* * * * *	1st + 3st + 1st
R2	* _ _ _ *	1st + 3sp + 1st
R3	* _ _ _ *	1st + 3sp + 1st
R4	* * * * *	1st + 3st + 1st

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→ 5 stars

① Outer loop (rows)
(1 to n)

② Inner loop (each row)

cout << "*" ; // First

for (1 to n-1) {
1st or last → "*" } ② work
else → ""

cout << "*" ; // Last



$n=4$

Inverted & Rotated Half-Pyramid

— — — *	R1	3 sp	+ 1 st
— — **	R2	2 sp	+ 2 st
— ***	R3	1 sp	+ 3 st
****	R4	0 sp	+ 4 st

↓

$i=1$	$n-i$ spaces
$i=2$	$4-1 = 3$
$i=3$	$4-2 = 2$
$i=4$	$4-3 = 1$
	$4-4 = 0$

① outer loop (rows)
(1 to n)

② inner loop (each row)

a) Spaces (1 to $n-i$)
`cout << " " ← work`

b) Stars (1 to i)
`cout << " " ← work`

`cout <<`

 $n=4$

code.cpp

code.cpp > main()

```
6
7     for(int i=1; i<=n; i++) {
8         //spaces
9         for(int j=1; j<=n-i; j++) {
10             cout << " ";
11         }
12
13         //stars
14         for(int j=1; j<=i; j++) {
15             cout << "*";
16         }
17         cout << endl;
18     }
19     return 0;
```

PORTS

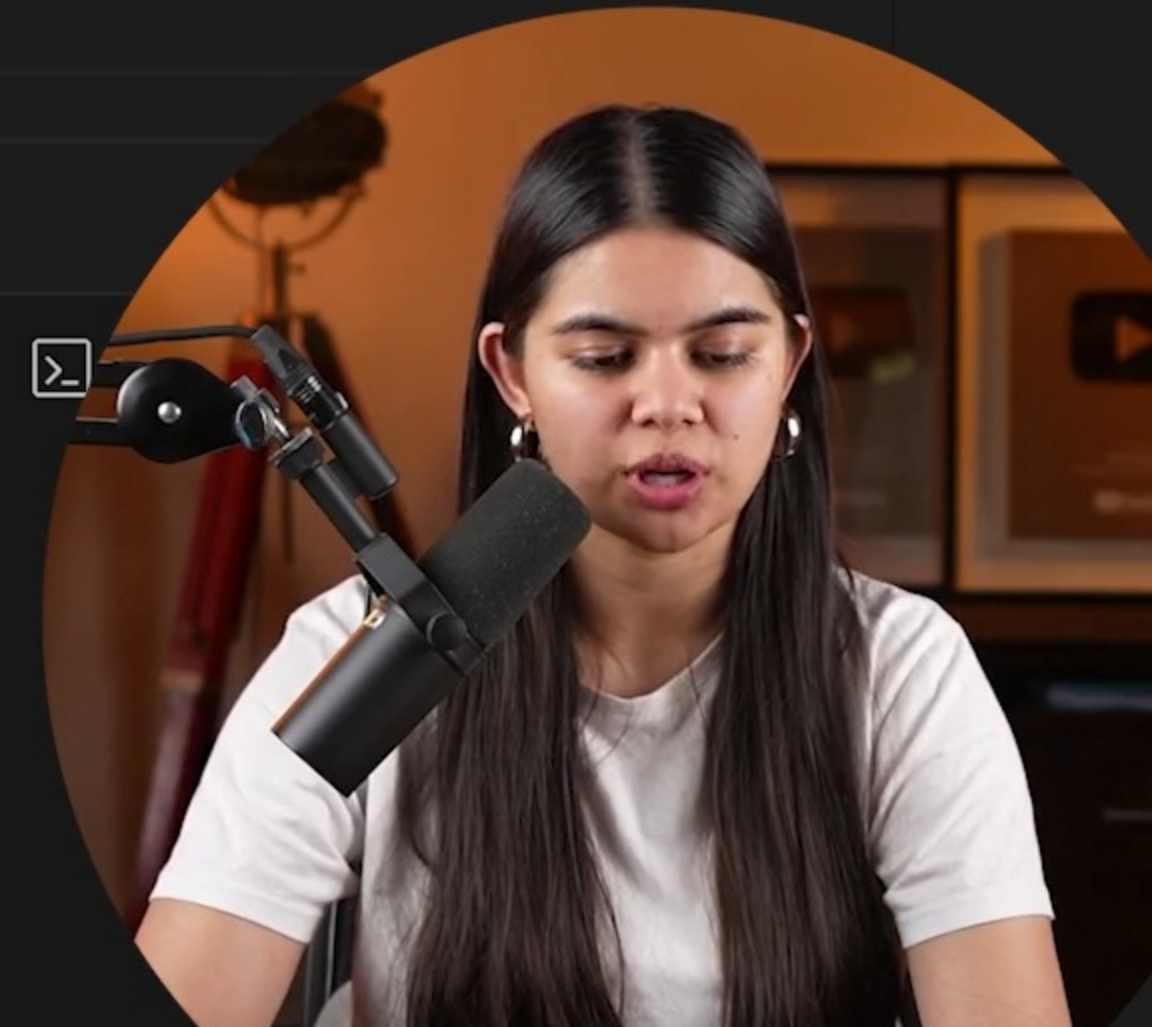
PROBLEMS

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apnacollege@Amans-MacBook-Pro Cpp Codes %



Print Floyd's Triangle

$n=5$
||

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

① outer loop (



Print Floyd's Triangle

R1 1
R2 2 3
R3 4 5 6
R4 7 8 9 10
R5 11 12 13 14 15

1 el

2 el

3 el

4 el

5 el

i th

i times
(1 to i)

num = 1

① outer loop (rows)
(1 to n)

② inner loop (each row)
elements
(1 to i)

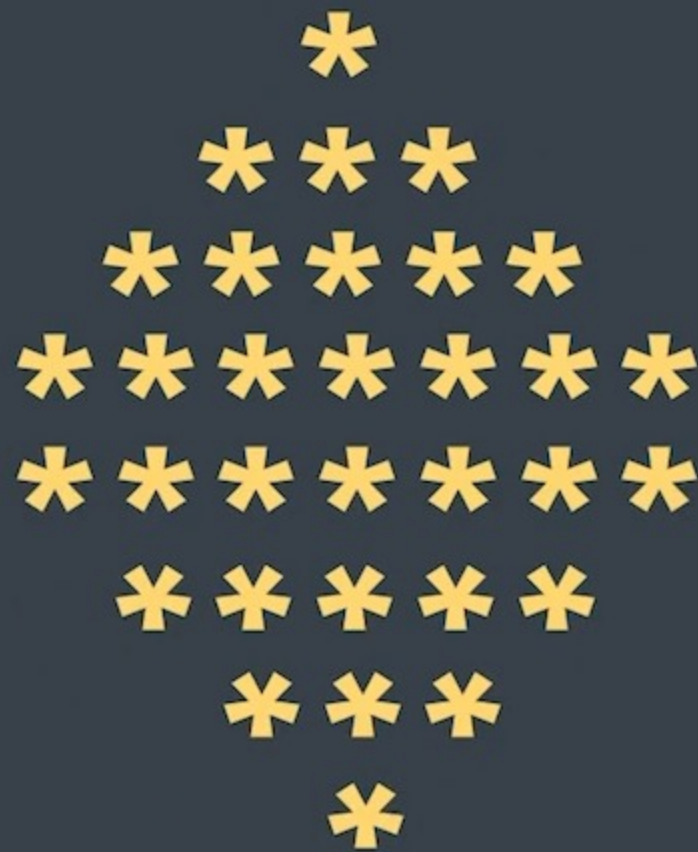
③ work?

cout << num;
num++;

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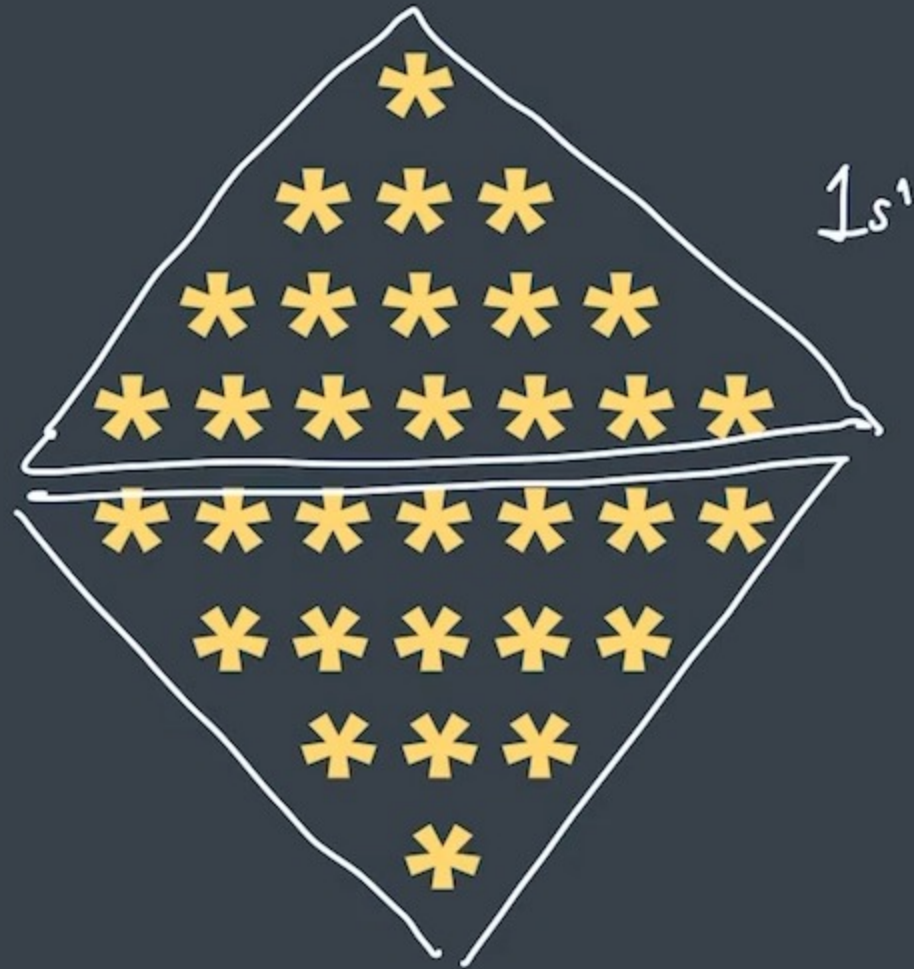
Diamond Pattern



$n = 4$



Diamond Pattern

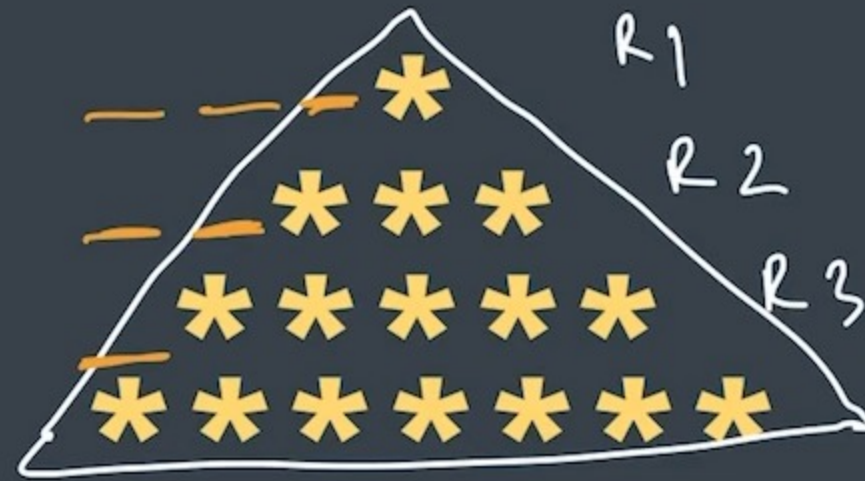


$n = 4$



Diamond Pattern

$n=4$



$3sp + 1st$
 $2sp + 3st$
 $1sp + 5st$
 $0sp + 7st$

1st pyramid

① outer loop (rows)
(1 to n)

② inner loop (each row)
elements

a) (1 to $n-i$) $cout << " "$
 b) (1 to $2*i-1$) $cout << "*"$

$i=1 \rightarrow 1$	$2*i-1 = 1$
$i=2 \rightarrow 3$	$2*i-1 = 3$
$i=3 \rightarrow 5$	$2*i-1 = 5$
$i=4 \rightarrow 7$	$2*i-1 = 7$



$n=4$

code.cpp ×

code.cpp > main()

```
4  int main() {
5      int n = 4;
6
7      //1st pyramid
8      for(int i=1; i<=n; i++) {
9          //spaces
10         for(int j=1; j<=n-i; j++) {
11             cout << " ";
12         }
13         //stars
14         for(int j=1; j<=2*i-1; j++) {
15             cout << "*";
16         }
17         cout << endl;
18     }
19
20     //2nd pyramid
21     for(int i=n; i>=1; i--) {
```

PORTS

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```
*****
***
*
```

apnacollege@Amans-MacBook-Pro Cpp Codes %

⌕ 0 0 0

Ln 18, Col 6 Spaces: 4 UTF-8



Print Butterfly Pattern



```
*           *  
**        **  
***     ***  
*****  
*****  
***     ***  
**        **  
*           *
```

n = 4



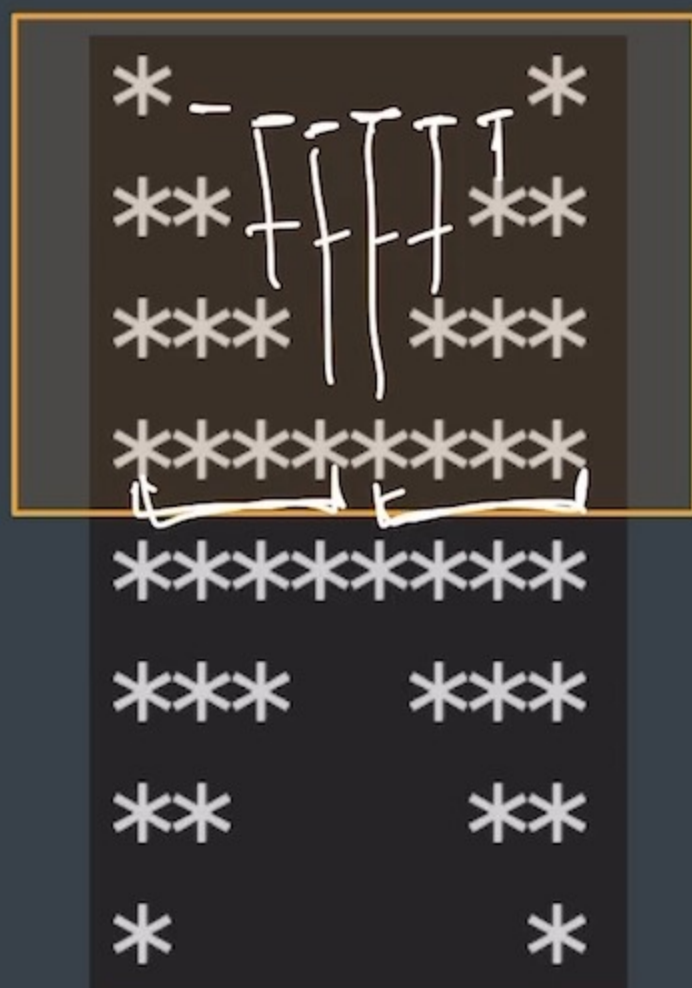
Print Butterfly Pattern

$n=4$



Pattern1

- ① outer loop (rows)
(1 to n)
- ② inner loop



R1 : 1st + 6sp + 1st
 R2 : 2st + 4sp + 2st
 R3 : 3st + 2sp + 3st
 R4 : 4st + 0sp + 4st



$n = 4$

Print Butterfly Pattern

$n=4$

```

*      *
**     **
***    ***
****   ****
*****

```

```

* - - - *
** - - **
*** - ***
*****

```

R1 : 1st + 6sp + 1st
 R2 : 2st + 4sp + 2st
 R3 : 3st + 2sp + 3st
 R4 : 4st + 0sp + 4st



$2 * (n - i)$

Pattern1

① outer loop (rows)
(1 to n)

② inner loop

a) stars (1 to i)

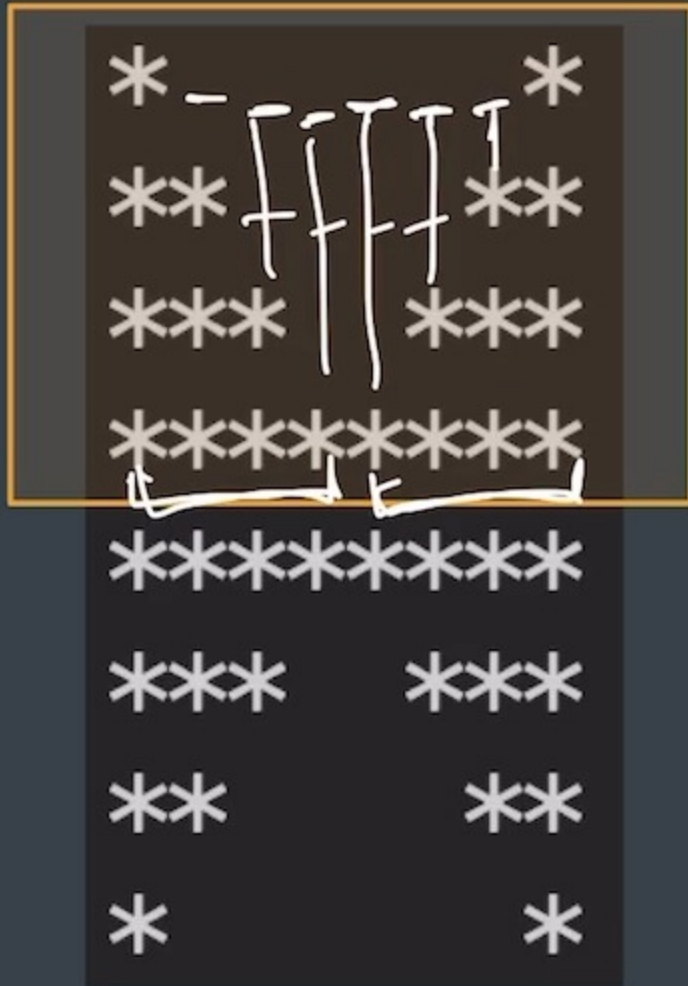
b) spaces

c) stars (1 to i)



Print Butterfly Pattern

n=4



R1 : 1st + 6sp + 1st
R2 : 2st + 4sp + 2st
R3 : 3st + 2sp + 3st
R4 : 4st + 0sp + 4st

$2 \times (n - i)$

Step 1

① outer loop (rows)
(1 to n)

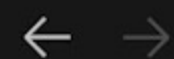
② inner loop

a) stars (1 to i)

b) spaces (1 to $2 \times (n - i)$)

c) stars (1 to i)

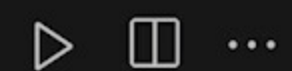




Cpp Codes



APNA
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code.cpp x



code.cpp > main()

```
4 int main() {  
5     int n = 4;  
6  
7     //upper half  
8     for(int i=1; i<=n; i++) {  
9         for(int j=1; j<=i; j++) {  
10             cout << "*";  
11         }  
12         for(int j=1; j<=2*(n-i); j++) {  
13             cout << " ";  
14         }  
15         for(int j=1; j<=i; j++) {  
16             cout << "*";  
17         }  
18     }  
19 }
```

PORTS

PROBLEMS

OUTPUT

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zsh +

apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out

```
*      *  
**     **  
***    ***  
*****
```

apnacollege@Amans-MacBook-Pro Cpp Codes %



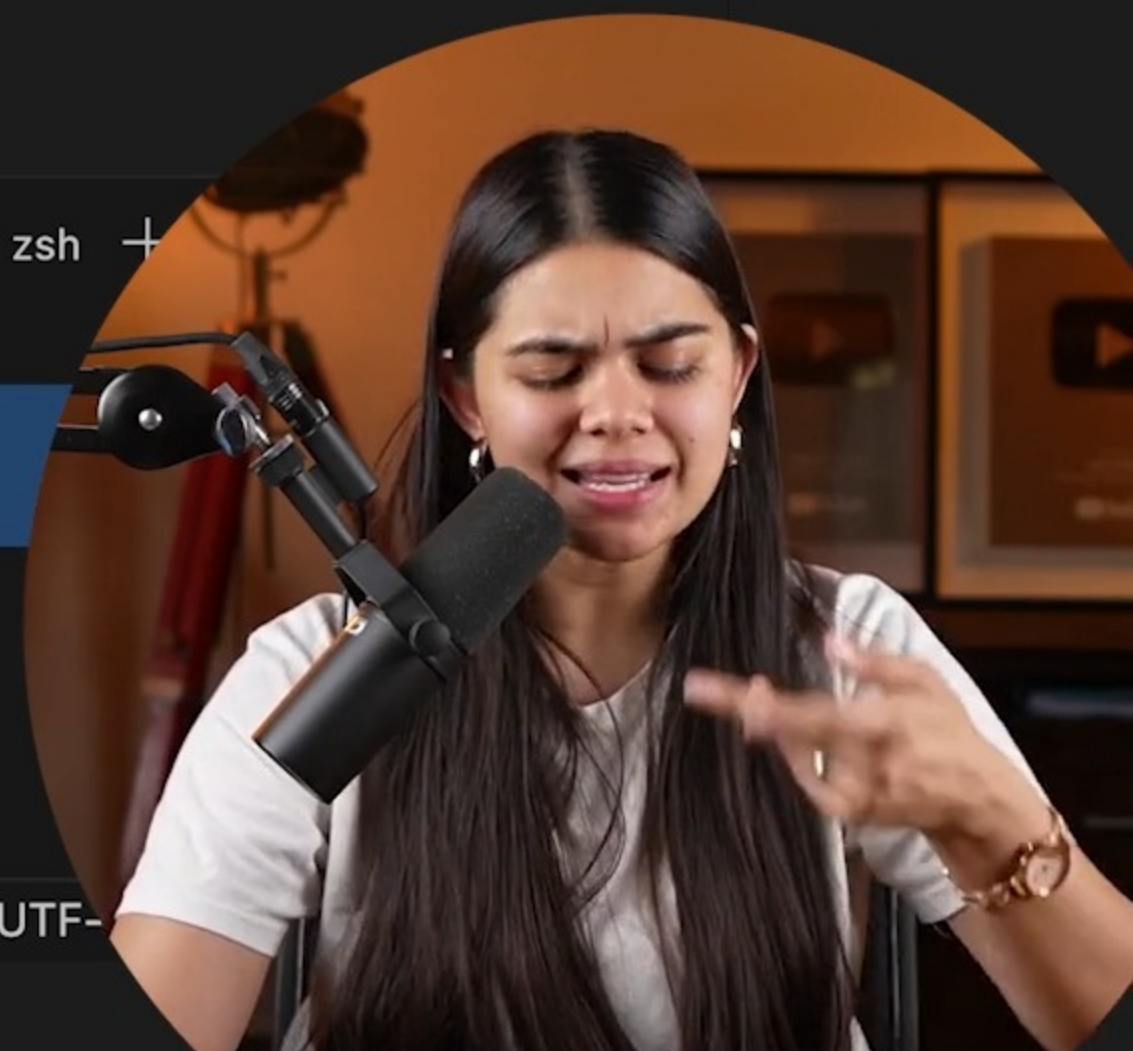
< 0 0 0



Ln 5, Col 14

Spaces: 4

UTF-



Print Butterfly Pattern

$n=4$

```

*      *
**     **
***    ***
****   ****
*****
****   ****
***    ***
**     **
*      *
  
```

Pattern

① outer loop (rows)
(n to 1)

② inner loop

a) stars (1 to i)

b) spaces (1 to $2 \times (n-i)$)

c) stars (1 to i)

```

*      *
**     **
***    ***
*****
*****
***    ***
**     **
*      *
  
```

R4 4 st
R3 3 st
R2 2 st
R1 1 st

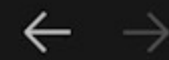
0 sp 4 st
2 sp 3 st
4 sp 2 st
6 sp 1 st

$$2 \times (n-i) = 2 \times$$

$$i=2$$



$n=4$



code.cpp ×

code.cpp > main()

```
8   for(int i=1; i<=n; i++) {
9       for(int j=1; j<=i; j++) {
10          cout << "*";
11      }
12      for(int j=1; j<=2*(n-i); j++) {
13          cout << " ";
14      }
15      for(int j=1; j<=i; j++) {
16          cout << "*";
17      }
18      cout << endl;
19  }
20  return 0;
21 }
```

PORTS PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

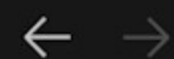
zsh +

● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out

```
*      *
**     **
***    ***
*****
```

○ apnacollege@Amans-MacBook-Pro Cpp Codes %

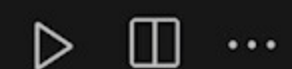




Cpp Codes



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code.cpp

code.cpp > main()

```
21 //lower half
22 for(int i=n; i>=1; i--){
23     for(int j=1; j<=i; j++) {
24         cout << "*";
25     }
26     for(int j=1; j<=2*(n-i); j++) {
27         cout << " ";
28     }
29     for(int j=1; j<=i; j++) {
30         cout << "*";
31     }
32     cout << endl;
33 }
34 return 0;
```

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PORTS

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OUTPUT

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clang +

apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out

```
*      *
**     **
***    ***
*****
```

apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out



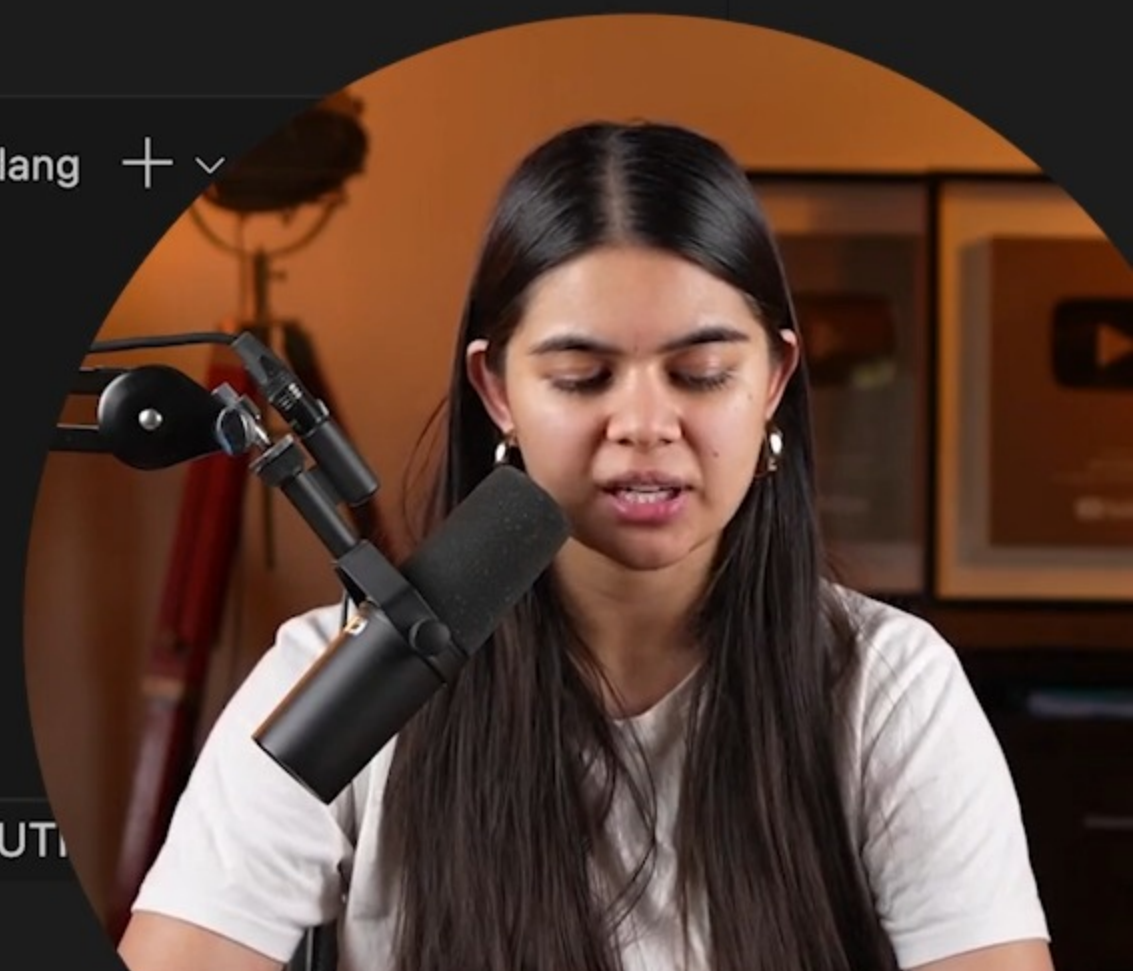
< 0 0 0

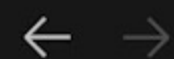


Ln 22, Col 30 (26 selected)

Spaces: 4

UT





Cpp Codes



APNA
COLLEGE



code.cpp x



code.cpp > main()

```
21 //lower half
22 for(int i=n; i>=1; i--){
23     for(int j=1; j<=i; j++) {
24         cout << "*";
25     }
26     for(int j=1; j<=2*(n-i); j++) {
27         cout << " ";
28     }
29     for(int j=1; j<=i; j++) {
30         cout << "*";
31     }
32     cout << endl;
```

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PORTS

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DEBUG CONSOLE

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zsh + v [icon] [icon] ...

apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out

```
*      *
**     **
***    ***
*****
```

apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out

```
*****
***    ***
**     **
*      *
```

apnacollege@Amans-MacBook-Pro Cpp Codes %



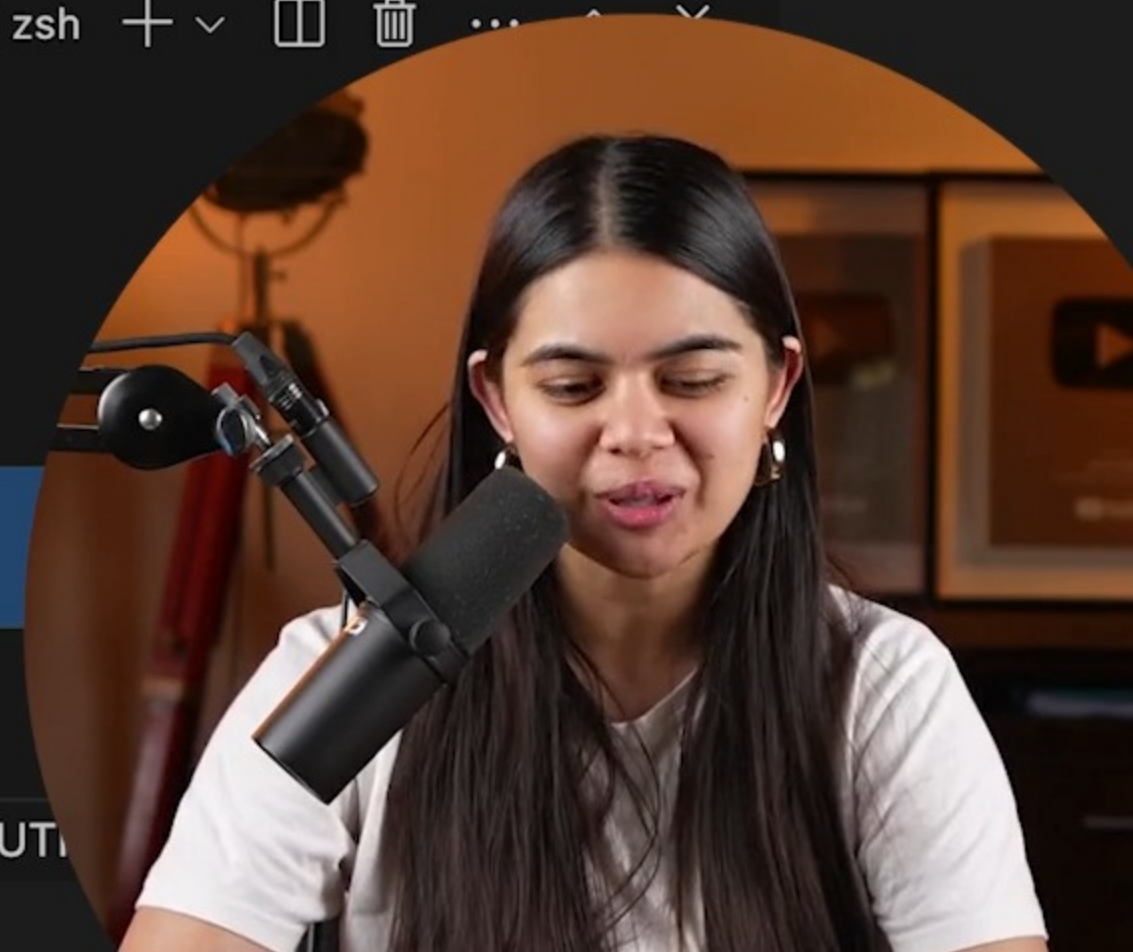
< 0 0 0



Ln 22, Col 30 (26 selected)

Spaces: 4

UT



13:00 / 14:53

